DEGRACE companions in MG-GLAM: status

What landed, how it was verified, what the two pilots say, what is left

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17 October 2026 merged into `cz`; plan in `docs/`

One week from plan to merged code

- **Scope** (plan of 16 Oct): 42 exact outputs per box (17 primary redshifts + 25 stations at $\Delta\chi = 256\,h^{-1}$Mpc), lens planes instead of snapshots, and the extra products of the brief: δ_L, low-k density and momentum modes, a hashed subsample, f_R at halo centres.
- **Implemented** on `degrace-track-a` (17 commits, 53 files, +10.8k lines): items I1–I10; I11 (optional) not done.
- **Verified**: 180-check regression suite + four unit/integration scripts, six independent code reviews (0 blockers, 8 majors, all fixed), two pilots on ECOSMOG node 2.
- **Merged** into `cz` today (fast-forward, suite green on `cz`).

Take-away

Everything a companion run needs exists, is tested, is on the mainline and has run twice at 1024^3. Three design decisions and one $0.5\,\%$ question remain.

Status of the plan items

item	what	status	evidence
I1	exact-z scheduler accepts arbitrarily close output pairs	**done** (`cz` ae4fa9d)	suite case 7
I2	lens planes: 12 per output, $\Sigma/\bar{\Sigma}$ on 2048^2	**done**	L1–L3
I3–I5	input generator, `OutputSchedule.dat`, per-output classes	**done**	cases 9, 4e; S2 (42 epochs = ECOSMOG to 10^{-7})
I6	provenance per box	partial	JSON lacks generator/LPT/seed (minor)
I7	$\delta_L(k)$ dump, route 1: in our FML IC generator	**done**	V3, V7, V8; `run_deltaL_test.sh`
I8	low-k δ_m and $(1+\delta)$v modes	**done**	V1, V2, mode-by-mode
I9	ID-hashed quantised subsample, 1 in M	**done**	V5, V6; `run_ic2pm_test.sh`
I10	f_R at BDM halo centres	**done**	unit test + f(R) smoke case
I11	optional extras	**open**	not started
pilot	node 2 ΛCDM, $N_g = 2048$	**done**	verified, 0 FAIL
pilot	node 2 f(R), extras + subsample + f_R	**done**	verified, 0 FAIL; 6 h 08 min

All flags default off: without the new `Init.dat` lines the code follows the `cz` path (review 2), apart from `OutputSchedule.dat` and an appended `Setup.dat` block.

Schedule and output classes (I1, I4, I5)

- `#outputs = -42`: every requested epoch is hit exactly (a step moved onto it or inserted); pairs closer than 0.5 % allowed (four in the DEGRACE list).
- Pilot: 181 steps = 163 ladder + 18 inserted, $\mathrm{d}a_0 = 6 \times 10^{-4}$ from $z = 49$.
- `OutputSchedule.dat`: step, a, z, χ, exact flag, class.
- Class 1 = full (halo finder + extras), 2 = station (planes + $P(k)$). Production decision: extras at all 42 outputs.

step	a	z	χ	exact	class
57	0.10275	8.732	8344.5	1	1
63	0.11832	7.452	8104.7	1	2
105	0.33333	2.000	5190.2	1	1
127	0.50000	1.000	3306.9	1	1
181	1.00000	0.000	0.0	1	1

χ in h^{-1}Mpc. Epochs equal the pinned ECOSMOG list of node 2 to 10^{-7}.

Lens planes (l2): cheap, verified, 1 % of the run

- Per output: 3 axes $\times$ 4 slabs of $256\,h^{-1}$Mpc, mean of $1+\delta$ over the slab column, block-averaged to 2048^2; float32, 201 MB per output, 8.5 GB per box.
- `.meta` sidecar: geometry, epoch, pixel convention, cosmology, header.
- Cost on the pilot: 1.8 s per output; the 201 MB write 12 s.

check	result
L1 mass conservation per axis	10^{-7}
L2 vs `ProjectDENSIT`	10^{-5}
L2 36^2 = block mean of 72^2	10^{-5}
L3 P_{2D} vs projected P_{3D}	3σ per ring; 3–7 %

Open question for Baojiu

Plane normalisation ($\Sigma/\bar{\Sigma}$ vs Σ); do the stations also need halo catalogues?

Fourier dumps, subsample, f_R at haloes (l8–l10)

product	what	verified
`deltam`	$\delta_k = N^{-3}\sum_x \delta(x)e^{-ik\cdot x}$, cube $\lvert k_i\rvert \le k_c = 1.5$, CIC on N_g, complex64 $C(0{:}n, -n{:}n, -n{:}n)$	V1 layout; V2 = `PowerDM.R` to 10^{-4}
`mom_x,y,z`	$(1+\delta)\,v$ on $N_{\rm dep} = N_g/2$, v at $a_v = a - \mathrm{d}a/2$, $k_c = 0.75$ (below $k_{N,\rm dep}/4$: no interlacing)	V6; mode by mode vs decoded particles 7×10^{-5}
`Subsample`	keep 1 in M by splitmix64(ID $\oplus$ salt), int16 positions and velocities, deterministic across twins, threads, restarts	V5 (10^7 IDs), V6
`fR_halo`	$f_R = f_{R,\rm bg}(a_{\rm old})\,u^{n+1}$ at every BDM centre	unit test; f(R) smoke case

- Storage at 1024^3 per output: δ_m 0.47 GB, momentum 0.41 GB, subsample ($M = 100$) 0.13 GB.
- From the review: `ic2pm` places every particle at array index = Gadget ID and tags the lattice order (`IDz` 2LPTic, `IDx` FML) in the PM header, so the subsample index gives the Lagrangian position.

δ_L from the IC generator (I7, route 1)

- The FML generator writes `<prefix>_deltaL.bin/.meta`: its Gaussian field at $z_{\rm ini}$ divided by $D(a_{\rm ini})$, same layout and convention as `deltam`; $k_c = 1.5$, 0.47 GB.
- Lua: `write_deltaL`, `deltaL_kc`; on in the production input. Costs 1 s.
- Baojiu's ICs and the legacy pilots are out of scope (decision of 17 Oct).

check	result
Zel'dovich: $\delta_L = -ik\cdot\Psi(k)/D$	every mode to 10^{-3}
2LPT ICs	residual 8×10^{-3} (2nd order)
wrong axis / sign / half cell	off by $10^4\times$
V3: ratio to $D(a)$, $k \le 0.15$	0.9997–1.0012
V3: cross-correlation, $k \le 0.05$	0.99998
V3: coordinate offset per axis	$\le 0.03\,h^{-1}$Mpc
V7: 1/2/4 MPI ranks	identical md5

Lesson

CIC density of particles sitting on grid nodes is not linear in the displacement (phase scatter $\sim k\Delta/2$ per mode at early times): compare dumps with binned means and cross-correlations.

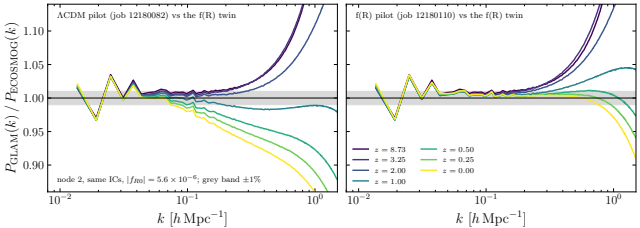
Pilot: node 2 ΛCDM, $N_g = 2048$ (job 12180082)

wall time	2 h 32 min, 64 cores, `cosma8-serial`
CPU efficiency	83 %
peak memory	118 GB (91 GB `PMP2main`)
steps	181, median 0.60 min
outputs	42 $P(k)$, 42 plane sets, 17 catalogues
disk per box	9.1 GB
core-hours	≈ 165 per ΛCDM box

step type	mean wall time
ordinary step	0.60 min
station output: 2048^3 $P(k)$ 25 s, planes 1.8 s, write 12 s	0.96 min
primary output: + BDM halo finder	2.6 min
all 42 outputs	43 min (29 %)
of which lens planes	$< 1\,\%$

- `check_run.py`: schedule, epochs, files, sizes, plane means, catalogues, ic2pm epoch: 0 FAIL.
- BDM dominates the output cost; the new products are cheap. The extras (modes, subsample, f_R) add at most 1.4 min per primary output on the f(R) pilot (two frames on).

Pilots vs the ECOSMOG twin: same ICs, mode-exact



- Left, ΛCDM vs the f(R) twin: flat $+0.5$ to $+0.7\,\%$ at $k < 0.1$ at every epoch, already at $z = 8.7$; late times: ECOSMOG above us by 5 % ($k = 0.1$–0.3) and 10 % ($k = 0.3$–1), the f(R) enhancement for $|f_{R0}| = 5.6 \times 10^{-6}$.
- Right, f(R) vs f(R), the real twin test: the same $+0.5$ to $+0.8\,\%$ at $k < 0.1$; within $\pm 1\,\%$ to $k = 0.3$ at every epoch; $-0.6\,\%$ at $k = 0.3$–1 at $z = 0$ (PM force resolution). $k > 0.3$ at $z > 3$: lattice discreteness.

The $+0.5\,\%$ at linear scales is the same in both gravity models

A code or ingestion offset, not physics: `ic2pm`'s half-step convention, POWMES normalisation or a 0.25 % growth difference; a $\mathrm{d}a_0 = 3 \times 10^{-4}$ restart of 20 steps isolates the half-step term.

f(R) pilot (job 12180110): completed and verified

	f(R), extras on (ΛCDM pilot)
wall, 64 cores	6 h 08 min (2 h 32 min)
CPU efficiency	83 % (83 %)
MaxRSS	177 GB (118 GB)
non-output step	1.72 min; multigrid 1.1 min (0.60 min)
station output	$+0.37$ min: P(k), planes ($+0.36$)
primary output	$+2.9$ min: BDM + extras ($+2.0$, BDM)
42 outputs	59 min = 16 % of the run (29 %)
disk per box	25 GB (9.1 GB)

- `check_run.py`: 42 exact outputs, 42 P(k) + planes, 17 catalogues, 0 FAIL, 0 WARN.
- Per primary: δ_m cube $n = 244$ (469 MB), momentum cubes $n = 162$ (3×138 MB), subsample of 10.74 M particles $= N/100$ to 0.03 %, no clipping (129 MB), f_R at 1.12 M haloes at $z = 1$ (9 MB): 1.0 GB.
- V9: the extras cost at most 1.4 min per primary (the excess over the ΛCDM primaries), ≤ 23 min $= 6\,\%$ of this run and $\leq 15\,\%$ of a ΛCDM run; an upper bound until `ExtraOutputs` gets per-product timers. Memory 177 GB on a 1 TB node.
- ≈ 390 core-hours per f(R) box at 1024^3.
- Caveats: $k_{c,\mathrm{mom}} = 1.0$ (use the momentum modes below $0.78\,h\,\mathrm{Mpc}^{-1}$); the binaries predate the review fixes.

Independent review: six reviewers, eight majors, all fixed

review	B	M	m	n
1 Fortran extras module	0	0	9	5
2 scheduler, parameters, build	0	2	6	6
3 particle subsample	0	1	4	6
4 δ_L generator dump	0	0	7	7
5 Python tools and tests	0	2	10	6
6 docs and provenance	0	3	11	8

Self-contained prompts and findings in `docs/reviews/degrace-track-a/`; rerunnable.

Majors and their fixes

- subsample "ID" was the IC file order $\rightarrow$ `ic2pm` scatters by Gadget ID and tags the lattice order
- optional `Init.dat` lines defaulted silently when malformed $\rightarrow$ stop with the line
- f_R gate on `MG_flag` only $\rightarrow$ also `MG_model = 1`
- twin script skipped early epochs (5-decimal header) $\rightarrow$ epochs from the schedule file
- no test sensitive to axis order/sign of the dumps $\rightarrow$ mode-by-mode test
- plan flags table, catalogue velocity epoch, interlacing status $\rightarrow$ documented, rules added

After the fixes: suite 180/180, four scripts green, merged.

Decisions pending and open items

Decisions (yours)

- **BDM catalogue velocities** are converted with $1/a$, the new products with $1/a_v$: a $1.1\,\%$ (first output) to $0.5\,\%$ ($z = 0$) offset. Fix the legacy finder (changes every GLAM catalogue) or keep documenting it (apply a_v/a offline).
- **Momentum deposit** at $N_g = 4096$ f(R): the transient $N^3_{\rm dep}$ buffer (34 GB) does not fit the 985 GB runs; reuse `FI` or restrict extras to $N_g = 2048$.
- **Interlacing**: not implemented; default $k_{c,\rm mom} = 0.75$ stays below $k_{\rm N,dep}/4$ and PMP2init refuses more. Implement if $k > 0.78$ momentum modes are wanted.
- **Disk**: 213 GB regenerable validation IC, 786 GB of February checkpoints in the high-res ΛCDM run, 459 GB in BDM validation work directories, 15 GB of superseded pre-fix fiducial catalogues.

Open minors (grouped in `TRIAGE.md`): parameter-path robustness, f_R product details, provenance JSON, test strength, docs drift. **Questions for Baojiu**: BDM cadence, plane normalisation, CAMB ω_c pitfall, r2g velocity convention, node 900.

Next steps

1. Explain the $+0.5\,\%$: the $\mathrm{d}a_0 = 3 \times 10^{-4}$ restart test (half-step term) and a POWMES normalisation check; add per-product timers to `ExtraOutputs` (the V9 split).
2. Settle the three decisions above; rebuild the frozen pilot binaries from `cz`.
3. Production recipe per node: `gen_inputs.py --node N --gravity {lcdm,fr}` $\rightarrow$ `submit.sh` (`ic2pm half` on Baojiu's ICs) $\rightarrow$ `check_run.py`; ≈ 165 core-hours per ΛCDM box, ≈ 390 for f(R); 9 GB (ΛCDM) or 25 GB (f(R) with the extras) per box.
4. Readers for the new products in `halocat` (planes, modes, subsample, f_R), following the `.meta` keys.
5. Reply to Baojiu with the cross-check of the 42 epochs and the open questions.

Summary

- The DEGRACE companion outputs (schedule, lens planes, δ_L, modes, subsample, f_R) are implemented, tested, reviewed and merged into `cz`.
- The ΛCDM pilot runs in 2.5 h per box with all 42 outputs, the f(R) pilot with all the extras in 6 h; the new products cost about 1 % of a run, the halo finder 23 %, the extras at most 6 %.
- Both pilots agree with the ECOSMOG twin to $+0.5$–$0.8\,\%$ at linear scales at every epoch, and the f(R) run reproduces the f(R) spectrum to $\pm 1\,\%$ up to $k = 0.3$; the small constant offset is common to both gravity models and is the one question still open.
- Three design decisions stand between here and production.